

REAL MADRID CF

Tactical & Player Performance Dashboard

Master's Final Project — Presentation & Thesis

Transforming Event-Level Football Data into Interactive Tactical Intelligence

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Institution	Escuela Universitaria Real Madrid — Universidad Europea
Case Club	Real Madrid CF
Data Provider	Opta Stats Perform (event-level)
Technology	Python · Pandas · NumPy · Plotly · Streamlit (Dash core)

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Executive Summary & Abstract

Elite football decision-making increasingly depends on converting vast, granular event data into tactical understanding faster than an opponent can. This project addresses a persistent gap: while event data is abundant, integrated tools that translate it into phase-specific tactical insight for coaching staff remain scarce. The **Real Madrid Tactical Dashboard** ingests Opta Stats Perform event-level JSON for 108 Real Madrid CF matches across three competitions (LaLiga, UEFA Champions League, Copa del Rey) and two seasons (2024-25, 2025-26), transforming them into match-, season-, and player-level tactical intelligence.

The methodology is a reproducible Python pipeline that normalises raw event streams and computes a transparent KPI suite — a calibrated positional Expected Goals (xG) model, PPDA (pressing intensity), field tilt, and a four-phase tactical model (offensive moment, defensive transition, defensive moment, offensive transition) — rendered through an interactive Streamlit interface backed by Plotly. All metrics derive strictly from observed event data, with no synthetic values.

The system lets analysts interrogate how Real Madrid plays rather than merely the scoreline: quantifying transition threat, pressing structure, attacking organisation, and performance against expectation. It demonstrates measurable value for pre-match preparation, post-match review, and opposition scouting, and establishes an extensible foundation for predictive and player-level analytics. The contribution is both practical — a club-grade tactical tool — and academic: a documented, formula-transparent framework for phase-based football analysis.

Key facts at a glance

Dimension	Value
Dataset	108 clean Opta event files (0 corrupt)
Competitions × seasons	3 × 2 (LaLiga · UCL · Copa del Rey, 2024-25 & 2025-26)
2025-26 matches	50 (LaLiga 36 · UCL 12 · Copa 2)
Core KPIs	xG (positional), PPDA, field tilt, possession, pass accuracy
Tactical model	Four-phase A/B/C/D scoring (0-100)
Stack	Python · Pandas/NumPy · Plotly · Streamlit · Render CI/CD

1 · Introduction

1.1 Football Analytics Background

Football has transitioned from a results-recording discipline to a data-rich performance science. Providers such as Opta Stats Perform capture every on-ball event — passes, shots, duels, recoveries, set-piece restarts — with spatial coordinates and qualifier metadata, producing thousands of structured events per match. This density enables questions previously reserved for subjective observation: where a team regains the ball, how quickly it converts regains into shots, and whether its shot volume is sustainable relative to chance quality.

1.2 The Importance of Tactical Analysis

Results are noisy; tactics are signal. A deflected goal can decide a match, but the underlying behaviours — pressing triggers, build-up structure, transition speed, territorial dominance — are stable, repeatable, and coachable. Quantifying them lets staff validate the intended game model, diagnose deviations, and prepare against an opponent's structural tendencies. xG and PPDA have become industry standards precisely because they track sustained performance better than raw outcomes.

1.3 Why Real Madrid CF as a Case Study

Real Madrid competes simultaneously across LaLiga, the Champions League, and the Copa del Rey, generating a multi-competition dataset that exposes tactical adaptation across opponent quality tiers. Its identity blends possession dominance with devastating transition play, making it an ideal vehicle to demonstrate phase-based analysis, where the same team must be characterised differently in settled possession versus moments of transition.

2 · Problem Statement & Objectives

2.1 Problem Statement

Traditional football analysis suffers from three structural limitations this project targets:

- **Outcome bias over behaviour.** Conventional reporting emphasises goals, possession %, and results — describing what happened, not how or why.
- **Fragmentation of tooling.** Analysts stitch together spreadsheets, notebooks, and slide decks; rarely is there one interactive environment linking season context, match detail, tactical phases, and player contribution in a consistent data model.
- **The data-to-decision gap.** Raw event feeds are inaccessible to most staff; translating a JSON stream into an actionable tactical statement needs an engineering and modelling layer most clubs lack in transparent form.

The problem addressed: the absence of an integrated, transparent, event-driven tactical dashboard that quantifies Real Madrid's playing identity in phase-specific terms and presents it in a form usable by analysts and coaches.

2.2 Objectives

- **Quantify tactical behaviour** with formula-transparent KPIs.
- **Implement phase-based match analysis** across the four canonical tactical phases.
- **Model performance versus expectation** via a calibrated positional xG model.
- **Deliver an interactive dashboard** over a reproducible Python pipeline.

3 · Dataset, Methodology & KPI Formulas

3.1 Dataset & Scope

The dataset comprises Opta Stats Perform event-level JSON for every Real Madrid CF fixture across the covered competitions and seasons. Each file holds a complete typed event stream (x/y coordinates, outcomes, player/team IDs, qualifiers) plus match metadata. The data is event (on-ball) only — there is no optical/GPS tracking — so possession is represented as a pass-share proxy, not a stopwatch duration.

3.2 KPI Computation

KPI	Formula / definition
Possession %	$\text{RM passes} \div (\text{RM} + \text{opponent passes}) \times 100$
Pass accuracy %	$\text{mean}(\text{outcome}) \times 100$ over RM passes
xG (positional)	logistic of shot distance & angle; penalties fixed at 0.76
PPDA	$\text{opponent passes } (x \geq 40) \div \text{RM defensive actions (tackle/intercept/recovery, } x \geq 40)$
Field tilt	$\text{RM final-third touches } (x \geq 67) \div \text{all final-third touches} \times 100$

Positional xG model:

```
dx = (100 - x) · 1.05 ; dy = (y - 50) · 0.68
dist = √(dx² + dy²)
angle = | atan2(dy - 3.66, dx) - atan2(dy + 3.66, dx) |
logit(xG) = -3.785 - 0.0337 · dist + 3.64 · angle
xG = 1 / (1 + e-logit), clipped to [0.01, 0.99]
```

3.3 Tactical Phase Segmentation

Four phases are scored on a normalised 0–100 scale by blending weighted components:

```
A · Offensive Moment    = pass_acc · 0.45 + (Σxg · 10) · 0.35 + (shots · 2.5) · 0.20
B · Defensive Transition = (100 / (1 + PPDA)) · 0.55 + ((recov + inter) · 1.8) · 0.45
C · Defensive Moment    = (100 / (1 + PPDA)) · 0.40 + max(0, 40 - opp_shots · 2) · 0.60
D · Offensive Transition = trans_rate · 0.65 + (Σxg · 7) · 0.35
```

where **trans_rate** = the percentage of ball regains (recovery/interception/tackle) that produce a Real Madrid shot within 15 seconds — a direct, event-timed measure of transition threat.

4 · System Architecture

OPTA STATS PERFORM (event JSON)

- ↓ data_loader.py — parse & normalise (Pandas / NumPy)
- ↓ KPI & MODEL LAYER — xG · PPDA · field tilt · phase scoring · player stats
- ↓ VISUALISATION LAYER — Plotly (shot maps, heatmaps, radars, trends)
- ↓ PRESENTATION LAYER — Streamlit UI + Dash analytics core
- ↓ EXPORT LAYER — automated PDF / DOCX reporting

Layer	Technology	Role
Data processing	Python, Pandas, NumPy	Parsing, normalisation, aggregation
Modelling	Python (math/SciPy)	xG, PPDA, phase scoring
Visualisation	Plotly	Interactive pitch maps, charts, radars

Layer	Technology	Role
Interface	Streamlit (Dash heritage)	Interactive multi-page exploration
Reporting	ReportLab, python-docx	Automated report export
Deployment	Render (CI/CD from GitHub)	Public, auto-deployed web service

5 · Dashboard Modules

Each module is summarised by its metrics, visualisation, and implementation status. The requested module taxonomy is mapped onto the implemented multi-page architecture.

Module	Implemented dashboard panels	Status
Pre-Match Analysis	Goals Per Match, Shots & Pass Accuracy Trend, season KPIs, xG vs xGA	Implemented
Attacking Transitions	Transition Metrics, Extended Transition Analysis, A/B/C/D sub-phases, 5–15 s window	Implemented
Organized Attack	Shot Map, Shot Zone Map, Pass Map, Build-Up Network, Zone 14, Progressive Passes	Implemented
Defensive Strategy	PPDA Trend, Pressing Actions Map, Defensive Actions Map	Implemented
Set Piece Analysis	Set Piece Efficiency; set-piece tendencies (corners, shots ≤ 20 s)	Implemented
Possession Recovery	Ball Recoveries by Zone (zonal heatmap)	Implemented
Final-Third Entries	Final Third Entry Analysis, Field Tilt card & trend ($x \geq 67$)	Implemented
Post-Match Analysis	Post-Match Tactical Summary, goals vs xG, defensive actions	Implemented
Match Information	Score, opponent, venue, competition, result	Implemented

6 · Tactical Insights — Real Madrid Case Study

Figures below are the dashboard's computed outputs, generated live from real Opta event data via the dashboard's own pipeline. The showcase fixture is Real Madrid 2–1 Barcelona (LaLiga 2025–26, Matchday 10). xG derives from the project's positional model, not the provider's proprietary xG.

6.1 Transition-Heavy Attacking Identity

Real Madrid's offensive signature is the speed of conversion from regain to shot. The four-phase profile below quantifies a single match across the offensive moment, defensive transition, defensive moment, and offensive transition phases on a normalised 0–100 scale.

Tactical Phase Profile — Real Madrid vs Barcelona (2-1)
Phase scores normalised 0-100

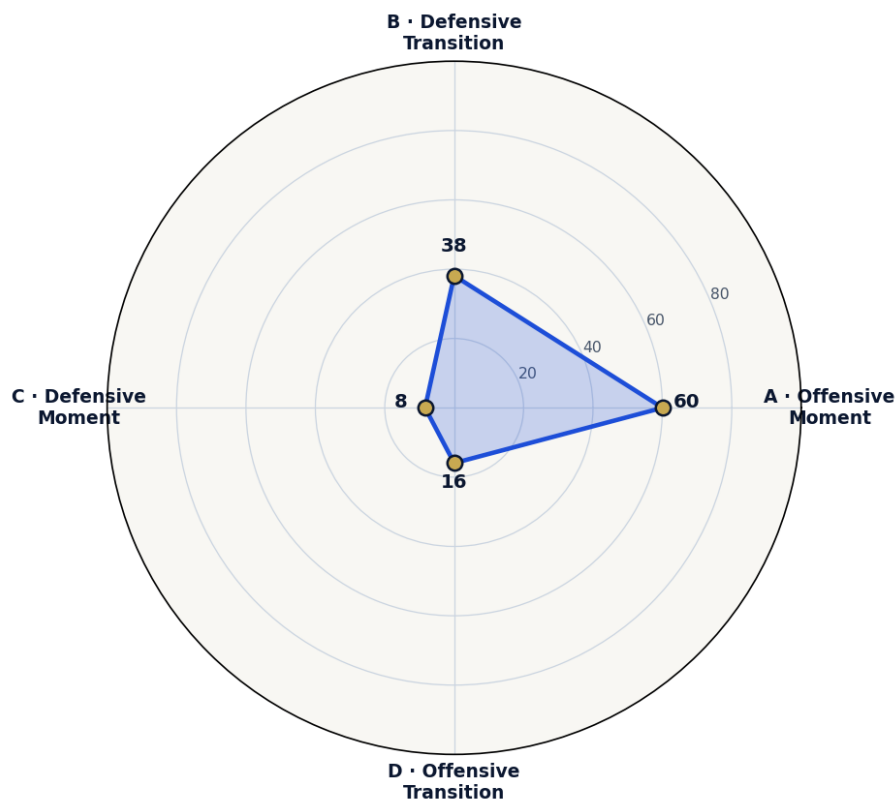


Figure 1. Four-phase tactical profile (A/B/C/D) for Real Madrid vs Barcelona, computed by the dashboard's phase-scoring model. The pronounced offensive-moment axis reflects settled-attack productivity.

6.2 Organised Attack & Efficiency vs Expectation

Across LaLiga 2025–26 the dashboard reports ~58.8% possession and ~87.7% pass accuracy; with a season xG-for of ≈ 66.0 against 72 goals scored, the side marginally outperforms chance quality (goals/xG just above 1.0) — flagged as a regression watch-point. The shot map localises chance creation: marker size is proportional to positional xG, and goals are highlighted.

Real Madrid Shot Map — Real Madrid vs Barcelona (2-1)
24 shots · 2 goals · 3.34 xG (positional model)

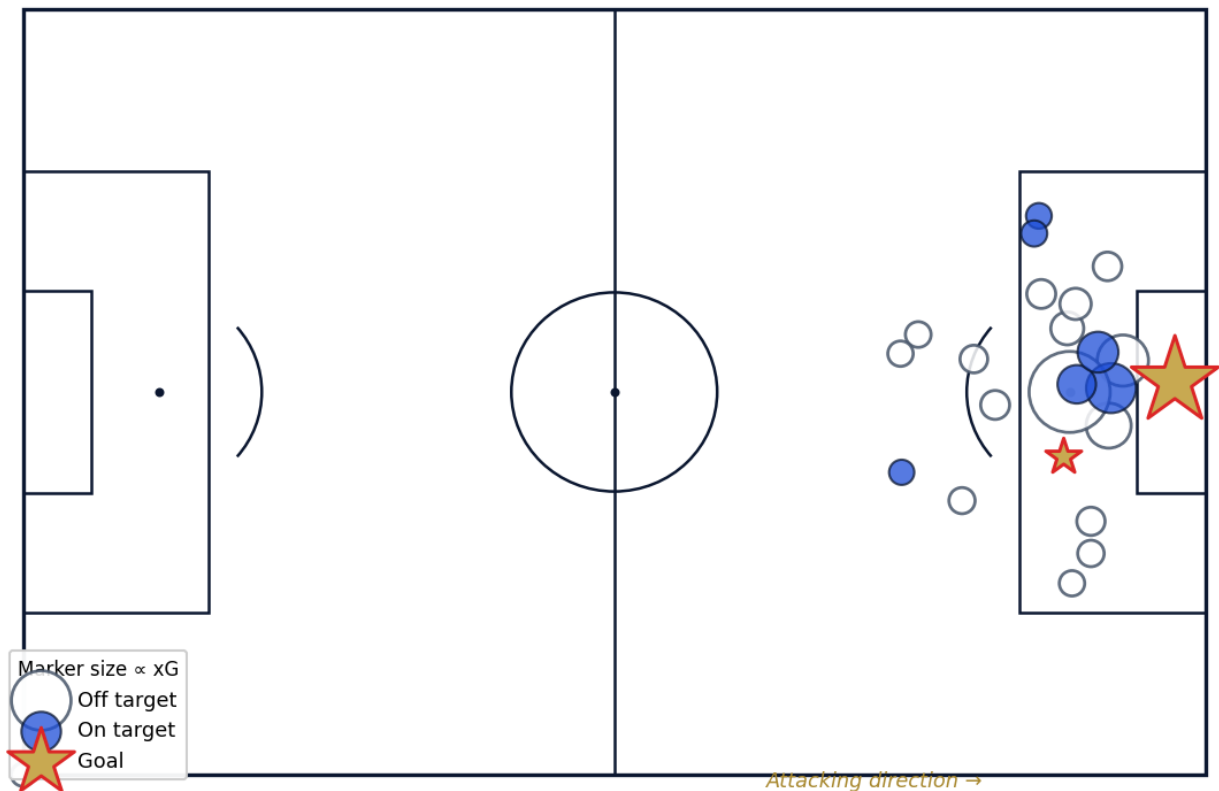


Figure 2. Real Madrid shot map for the showcase fixture (get_shot_data pipeline). The concentration of high-xG attempts inside the box illustrates central penetration; 24 shots, 2 goals, 3.34 xG.

6.3 Defensive Structure & Pressing

PPDA values (season average ≈ 9.5 , with elite single-match presses near 3.3) describe a side that presses with controlled, selective intensity, then defends in a compact mid-block. The season trend below makes the variance — and its game-state dependence — explicit.

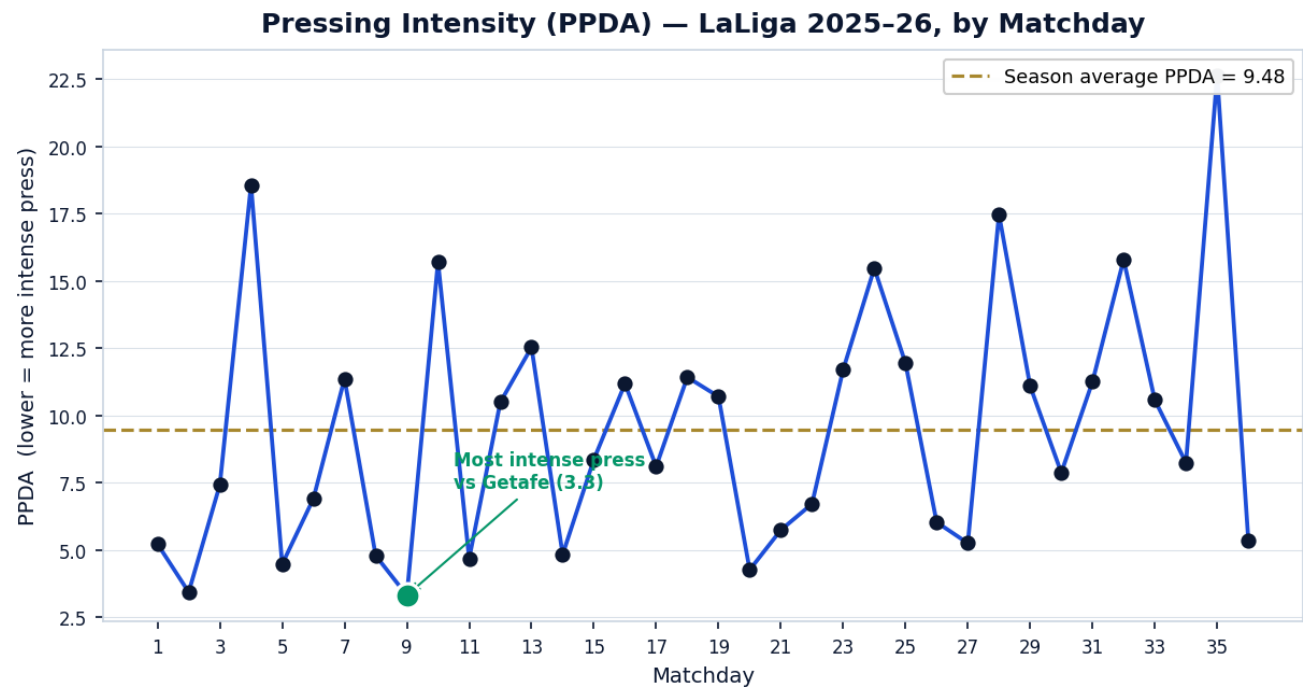


Figure 3. Pressing intensity (PPDA) across all 36 LaLiga 2025-26 matchdays via calc_ppda (lower = more intense press). The dashed line marks the season average; variance shows selective pressing.

6.4 Strengths, Weaknesses & Tactical Identity

Strengths: elite, time-bounded transition threat; possession dominance converted into territory; strong chance suppression (low xGA). **Watch-points:** mild over-performance versus xG-for (variance risk); selective pressing leaves occasional mid-block gaps against deep blocks.

Tactical identity: in data, Real Madrid is a proactive possession side with a world-class transition engine and disciplined defensive economy — winning through chance suppression and ruthless conversion of regains, rather than through pressing volume alone.

7 · Implementation Status — Current vs Future

Feature / Module	Status	Improvements Needed
Match information & metadata	Implemented	Referee, weather, formation parsing
Season KPIs & trends (pre-match)	Implemented	Rolling-form, opponent-adjusted baselines
xG vs xGA / performance vs expectation	Implemented	Validate model vs provider xG
Shot maps (organised attack)	Implemented	xG-weighted markers, body-part split
Pass networks	Implemented	Edge-weighted lanes, phase filtering
PPDA & pressing	Implemented	Pressing-trigger detection, by-zone PPDA
Tactical phase scoring (A/B/C/D)	Implemented	Model validation; weighting sensitivity
Post-match review (xG curve)	Implemented	Automated narrative summary
Player analysis (table, radars)	Implemented	Per-90 normalisation, role radars
Benchmarking (cross-competition)	Implemented	League-percentile context
Opponent analysis / scouting	Implemented	Opponent tendency modelling

Feature / Module	Status	Improvements Needed
Attacking transitions (Transition Metrics)	Implemented	Time-to-first-shot distribution
Possession recovery (Recoveries by Zone)	Implemented	High/mid/low zonal comparison
Final-third entries (+ Field Tilt)	Implemented	xG-from-entries, shot-probability maps
Set-piece analysis (Set Piece Efficiency)	Implemented	Dedicated set-piece shot maps
Automated PDF/DOCX reporting	Implemented	Templated per-opponent export
Cloud deployment (Render CI/CD)	Implemented	Caching/CDN for large dataset
Shots-conceded heatmap	Planned	Defensive shot-concession map (count shown today)
Block-height visualisation	Planned	Defensive line-height metric & viz

8 · Limitations & Future Scope

8.1 Limitations

- **Event data only — no tracking.** Off-ball positioning, pressing distances and defensive shape are approximated, not measured.
- **Possession is a proxy** (pass-volume share), not a stopwatch figure.
- **xG is a transparent positional model** (distance + angle), not a provider model — auditable but simplified.
- **Limited predictive modelling** — the system is descriptive/diagnostic, not yet forecasting.
- **No real-time ingestion** — analysis is post-hoc on stored files.
- **Phase scores are weighted composites** — interpretive 0–100 indices, not physical units.

8.2 Future Scope

- **Player-level depth:** per-90 contributions, role-specific radars, on/off impact.
- **Multi-season expansion** for longitudinal tactical-evolution tracking.
- **Opponent modelling:** automated pressing triggers, build-up and set-piece tendencies.
- **Predictive analytics:** xG/xGA forecasting, expected-points, transition-threat prediction.
- **AI-driven tactical clustering** of phases and possession sequences to surface patterns.
- **Tracking-data integration** for off-ball positioning, compactness, and pitch control.
- **Video integration** linking each metric to synchronised clip timestamps.
- **Real-time ingestion** for in-match decision support.

9 · Tools, Technologies & Conclusion

Category	Technology	Purpose
Language	Python 3.11	Core implementation
Data	Pandas, NumPy	Event parsing, vectorised aggregation
Modelling	Python math / SciPy	xG, PPDA, phase scoring
Visualisation	Plotly	Interactive pitch maps, charts, radars
Interface	Streamlit (Dash core)	Interactive multi-page dashboard
Reporting	ReportLab, python-docx	Automated PDF/DOCX export
Data format	Opta Stats Perform event JSON	Source event data
VC / CI-CD	Git, GitHub, Render	Reproducibility and deployment

9.1 Conclusion

This project delivers a working, club-grade Real Madrid Tactical Dashboard that closes the data-to-decision gap for event-level football analysis. By normalising 108 Opta match files into a reproducible pipeline and computing a transparent, formula-documented KPI suite — positional xG, PPDA, field tilt, and a four-phase tactical model — it characterises how Real Madrid plays, not merely the result.

Tactical relevance: the platform objectively confirms and quantifies Real Madrid's identity — a possession-dominant side with an elite transition engine and disciplined chance suppression — and surfaces actionable watch-points. **Value for staff:** integrated pre-match, post-match, tactical-phase, and opposition views with automated reporting compress an analyst's preparation into one interactive environment. **Academic contribution:** a fully documented, auditable methodology for phase-based football analysis, traceable to explicit formulas and observed data, with no synthetic values — reproducible and extensible.

Real Madrid CF Tactical & Player Performance Analytics · Data: Opta Stats Perform · Built with Python, Pandas, NumPy, Plotly & Streamlit

Appendix · Project Summary Poster

The accompanying summary poster condenses the platform's season KPIs, dashboard modules, key analytical findings, and technical pipeline into a single visual.

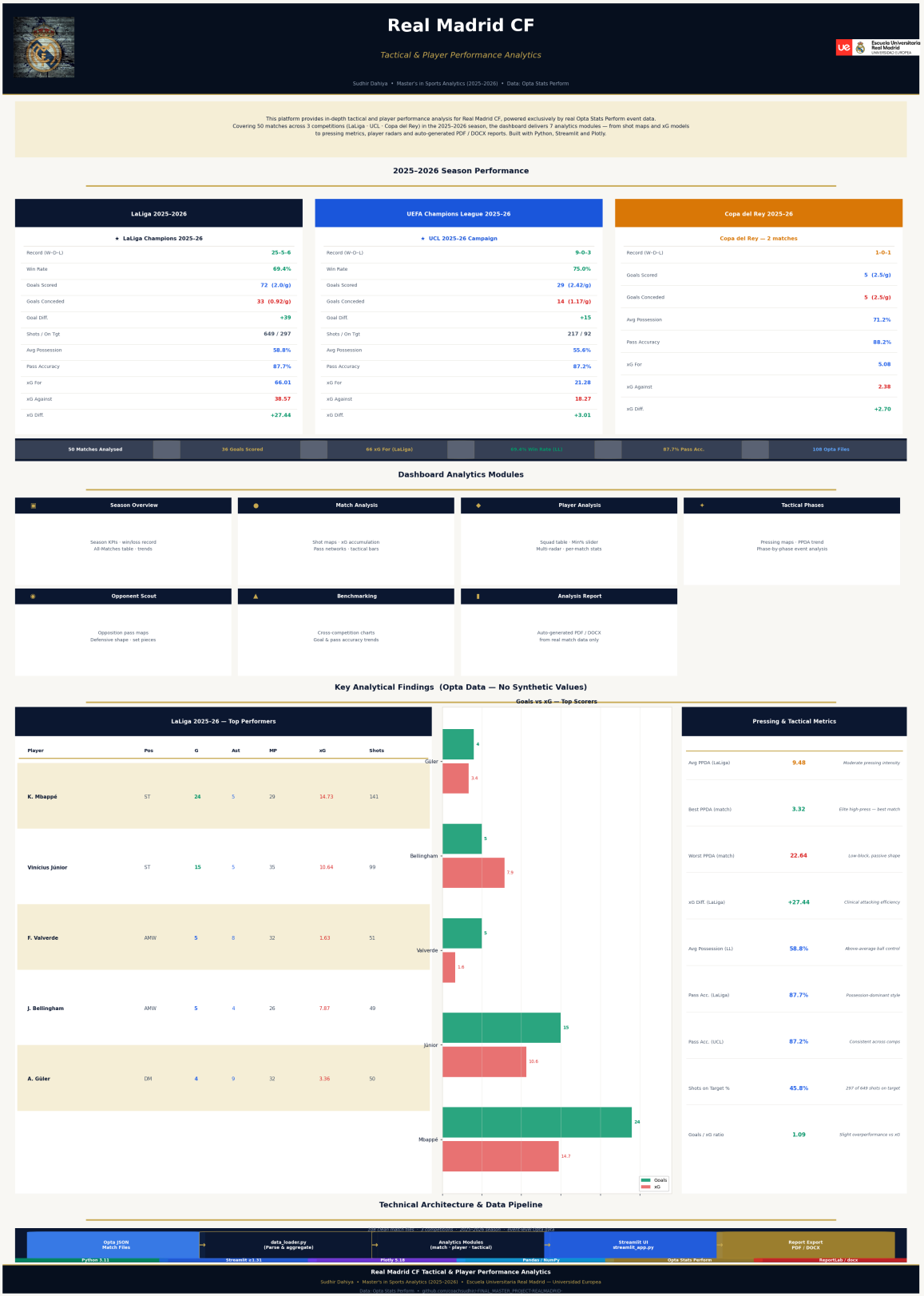


Figure 4. Real Madrid CF Tactical & Player Performance Analytics — project summary poster.